TECHNICAL SPECIFICATIONS

ResqBot - Catamaran Based Chassis

1. Executive Summary & Design Overview

The structural, dimensional, and material specifications for a lightweight, twin-hull (catamaran). Specifically for additive manufacturing using **PETG (Polyethylene Terephthalate Glycol).**

The catamaran-style configuration provides enhanced lateral stability in surface water conditions, while space-frame lattice arms distribute dynamic loading from the pontoon hulls into the central payload compartment.

2. Bounding Dimensions & Physical Envelope

Dimension Parameter	Value	Description / Reference
Overall Length (OAL)	**409.00 mm**	Maximum longitudinal profile along the pontoon axis
Overall Beam (Width)	**412.00 mm**	Total horizontal span from port to starboard hull outer edges
Overall Height (OAH)	**180.00 mm**	Total vertical height from keel base to tail fin apex
Lattice Arm Diagonal Span	**199.25 mm**	Structural truss frame length connecting hulls to central bay
Pontoon Mount Footprint	**110.90 × 52.00 mm**	Contact interface block for lattice arm pontoon mounting
Central Bay Footprint	**171.05 × 130.49 mm**	Top deck profile envelope for main payload compartment

3. Material & Structural Wall Thickness Profile

To optimize displacement and weight distribution while maintaining mechanical strength under wave impact, the chassis utilizes a differential wall thickness strategy:

SUB-ASSEMBLY ZONE	PETG WALL THICKNESS	SLICER / PRINT SPEC
Hinge Joints & Fins	4.0 mm - 5.0 mm	10+ Perimeters / Solid
Central Bay & Arms	2.5 mm - 3.0 mm	6-7 Perimeters
Pontoon Hulls	2.0 mm - 2.5 mm	5-6 Perimeters

3.1 Fabrication & Slicer Specifications

- **Primary Filament:** PETG (Polyethylene Terephthalate Glycol)
- **Pontoon Shells ($2.0--2.5\ \mathrm{mm}$):** Designed to minimize dead weight below the waterline while providing sufficient impact resistance against debris.
- **Central Bay & Lattice Arms ($2.5--3.0\ \mathrm{mm}$):** Higher wall density to absorb hull torsion during differential wave action and support internal payload mass.
- **Hinge Assemblies & Tail Fins ($4.0--5.0\ \mathrm{mm}$):** Near-solid printing required around pivot pin bores to handle shearing stress and dynamic tail fin drag.

4. Subsystem Layout & Internal Architecture

4.1 Pontoon Hulls (Port & Starboard)

- **Hull Geometry:** Hydrodynamic, low-drag displacement profile with a tapered bow.
- **Internal Cavity:** Hollow pontoon structure yielding an estimated internal volume of $700--800\ \mathrm{cm}^3$ per hull.
- **Reserve Buoyancy Integration:** High-density, custom-trimmed EVA foam blocks are inserted directly into the hollow pontoon cavities to guarantee positive flotation in the event of an outer shell breach.
- **Retention Method:** Secured via friction fitting or marine-grade silicone adhesive (RTV sealant). No internal structural recessed channels are needed.

4.2 Structural Lattice Arms

- **Truss Architecture:** Triangular space-frame geometry designed to distribute hydrodynamic drag evenly across the vessel frame.
- **Structural Span:** 199.25 mm per side, providing a wide 412 mm total beam for stability against capsizing.

4.3 Central Electronics Compartment

- **Envelope:** $171.05\text{ mm (L)} \times 130.49\text{ mm (W)}$.
- **Internal Component Integration:** All power distribution, logic boards, battery cells, sensors, and charging modules are fully enclosed within the compartment.
- **Ingress Protection Strategy:** The exterior of the central housing is printed solid on all sides (no front camera/sensor aperture cutouts) to protect against head-on wave splashes. Charging terminals and debug ports remain internal; access is conducted via the top hatch.

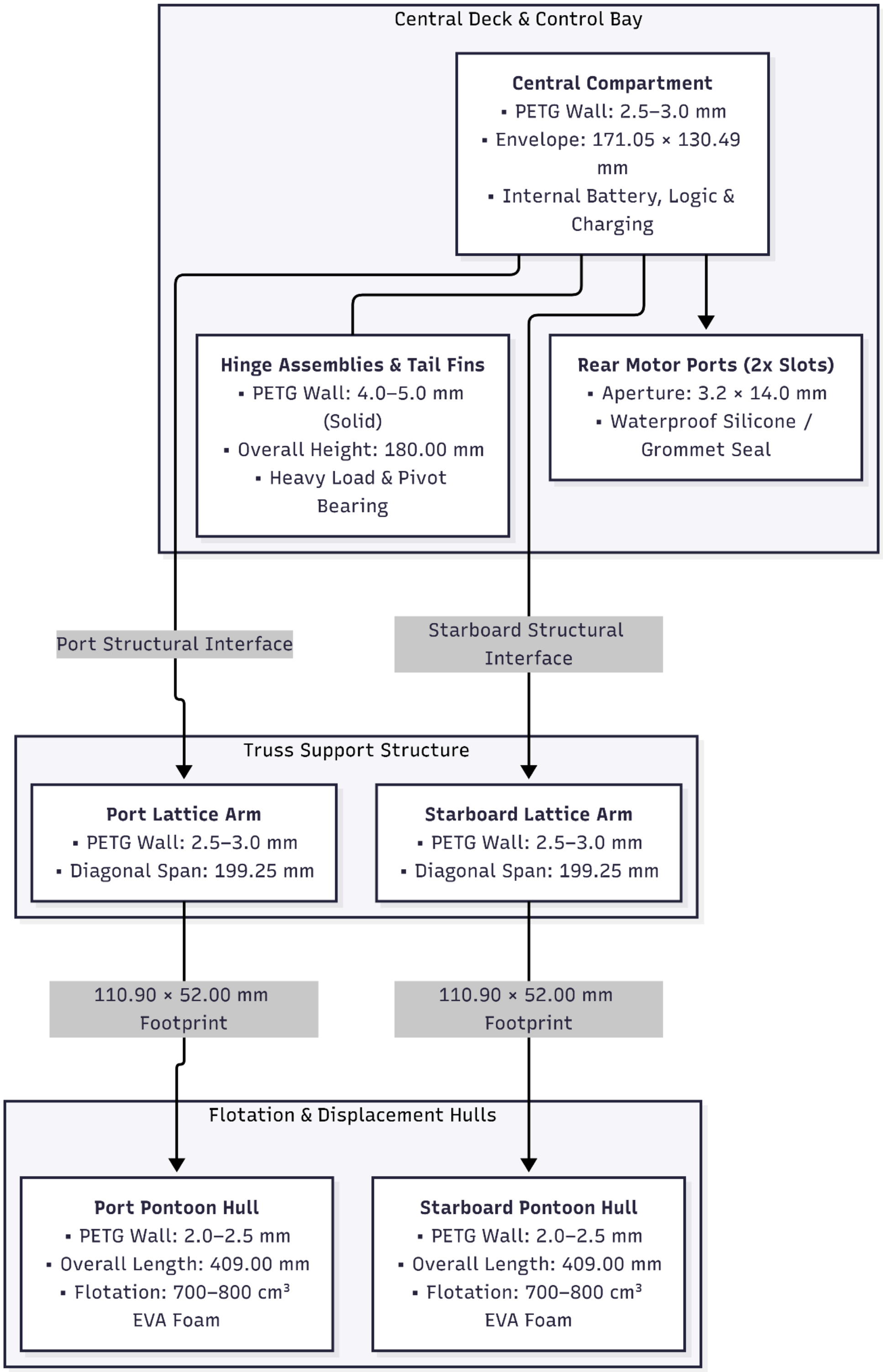
5. Port Geometry & Wiring Pass-Throughs

Port Designation	Location	Dimensions	Quantity	Primary Function
Motor Ports	Rear Face	$3.2\text{ mm} \times 14.0\text{ mm}$	2	Wire lead routing & propulsion motor mounts
Front Optics Port	Front Face	10 mm in diameter	2	“Eye” sockets for the camera (Thermal and Visual)
Front Middle Optic Port	Front Face	30 mm in diameter	1	Slot for the ultrasonic sensor right below the front optics to ensure close relative field of view.

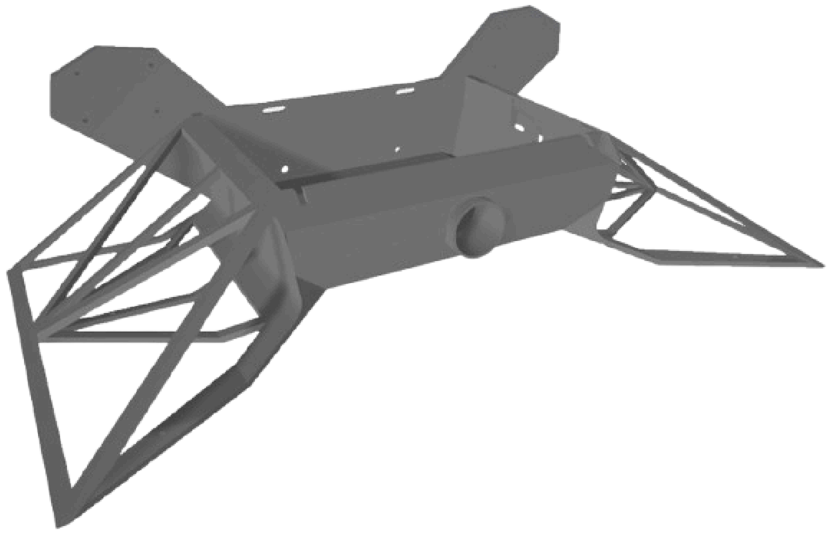
Sealing Protocol

To maintain watertight integrity, the rear $3.2\text{ mm} \times 14.0\text{ mm}$ motor ports must be sealed after wire harness installation using rubber cable grommets or **silicone seal potting compound.**

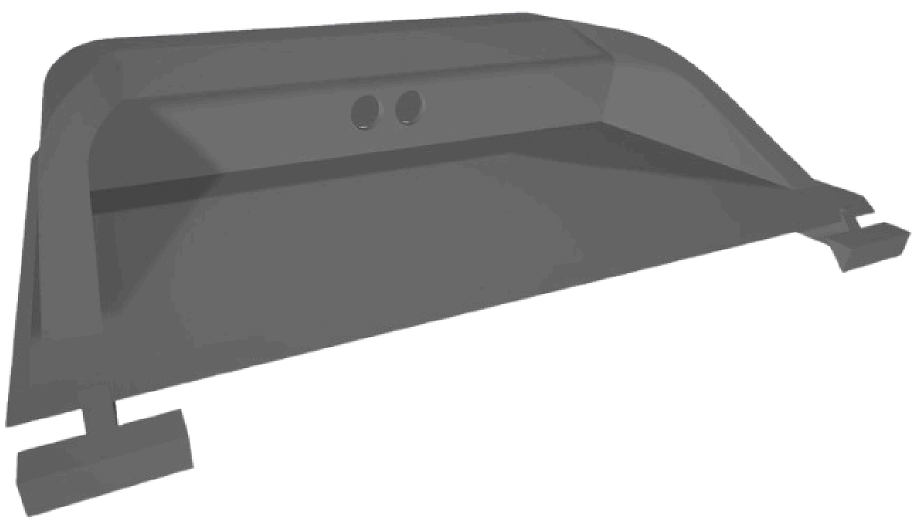
6. System Summary Matrix



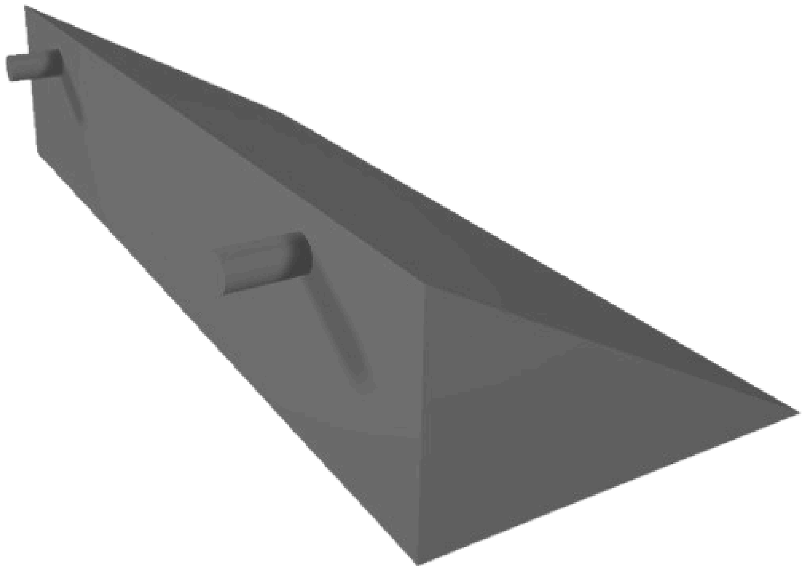
7. Complete Visual Presentation of Each Part



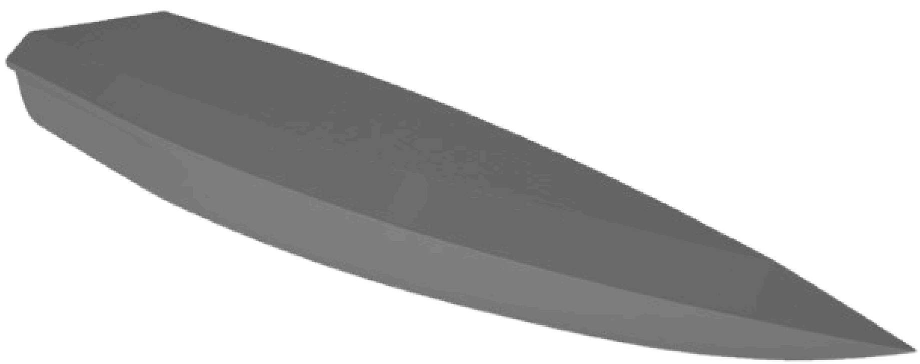
A. Compartment



B. Top Hatch / Lid



C. Motor Hinges



D. Left and Right Hull (Same Part) // Must be hollow from the inside as specified in earlier parts of the document.